Table of Contents

1.0 Project Overview & Objectives
2.0 Test Scope
3.0 Test Strategy
4.0 Test Schedule & Resources
5.0 Risk Analysis
6.0 Entry & Exit Criteria
7.0 Deliverables & Sign-off

1.0 Project Overview & Objectives

1.1 Project Background

The video streaming platform (https://www.youtube.com) is a comprehensive web application designed to facilitate the upload, sharing, and viewing of video content. This platform aims to provide a seamless and high-quality user experience for content creators and consumers alike, supporting a wide range of video formats and interactive features.

1.2 Testing Purpose

This testing effort, formally designated as Cycle 1 – Initial Testing, is being conducted to comprehensively assess the functional stability, performance, and usability of the video streaming platform, Version 1.0. The primary goal is to identify critical defects and ensure compliance with specified requirements prior to broader release.

1.3 Objectives

- Verify that users can successfully register, log in, and log out of the platform without errors.
- Validate the complete functionality of video upload, processing, and playback across supported devices and browsers.
- Ensure the accuracy and responsiveness of search and recommendation algorithms based on user queries and viewing history.
- Confirm the proper operation of user interaction features, including commenting, liking, and sharing videos.
- Assess the platform's performance under expected user loads for video streaming and data retrieval.
- Identify and document all critical and major defects, ensuring they are reproducible and clearly reported.
- Evaluate the user interface for intuitiveness and adherence to established design guidelines.

1.4 Quality Goals

- 95% of critical test cases must pass before the completion of Cycle 1 testing.
- No critical or major defects shall remain open at the conclusion of testing.
- Response times for key user actions (e.g., video playback start, search results) shall not exceed 3 seconds under specified load conditions.
- The application shall exhibit no crashes or unhandled exceptions during standard user workflows.
- The platform shall demonstrate compatibility with the latest stable versions of Chrome, Firefox, Safari, and Edge browsers.

2.0 Test Scope

Detected functionalities (from generated test cases)

Video Playback · Search · Channel Subscription · User Authentication · Video Upload · Liking and Disliking · Commenting · Sharing · Playlist Creation · Related Videos

2.1 In Scope

- User authentication (login, logout, account creation)
- Video playback functionality (play, pause, seek, volume control, resolution settings)
- Video search functionality (basic search, filters)
- Commenting and liking/disliking videos
- Subscription to channels
- Video uploading and management (for logged-in users)
- Playlist creation and management
- Homepage content display and recommendations
- Channel pages and content organization
- Responsive design across various viewports and devices
- Navigation across all public-facing pages
- Core user journeys including video consumption, content discovery, and interaction.
- Primary business flows related to content delivery and user engagement.

2.2 Out of Scope

- Backend administration tools, as these are internal applications not accessible to end-users.
- Internal APIs not directly exposed through the user interface, as these fall outside the scope of user-facing application testing.
- Load testing, as this is a specialized performance testing activity requiring different tools and methodologies beyond the current scope.
- Content moderation processes, as these often involve human review and separate internal systems.

2.3 Testing Types

Type	Description	Rationale
Functional Testing	Validation of all specified functionalities to ensure they perform as expected according to requirements.	To ensure all features of the YouTube web application operate correctly and meet user expectations.
UI/UX Testing	Assessment of the user interface for usability, aesthetics,	To guarantee an intuitive, consistent, and visually appealing

Type	Description	Rationale
	responsiveness, and adherence to design guidelines.	user experience across various devices and browsers.
Regression Testing	Re-execution of existing tests to confirm that recent code changes have not adversely affected existing functionalities.	To prevent unintended side effects from new features or bug fixes, ensuring stability of the application.
Smoke Testing	A quick run-through of the application's most critical functionalities to ensure the build is stable enough for deeper	To confirm fundamental functionalities are working post-deployment or major integration, allowing for efficient use of testing

2.4 Assumptions

- The test environment will be stable and reflect production configurations.
- Necessary test data will be available and prepared prior to test execution.
- Access to required user accounts (e.g., standard user, content creator) will be provided.
- All dependencies, such as network connectivity and third-party integrations, will be operational.
- Requirements and design specifications will be finalized and available for reference.

3.0 Test Strategy

3.1 Approach

The testing strategy for YouTube's web application will employ a multi-faceted approach, emphasizing comprehensive coverage and iterative feedback. This involves a planned sequence of testing activities, commencing with initial smoke tests, progressing through functional and UI-UX testing, and concluding with performance and regression cycles to ensure the application's stability and reliability across specified environments.

3.2 Test Levels

Level	Description	Responsible
Component Testing	Individual modules and features, such as video playback, comment sections, and user authentication, will be tested in isolation to verify their correct functionality and adherence to specifications.	Development Team
Integration Testing	Testing will focus on the interfaces and interactions between different components, ensuring seamless data flow and communication between features like search functionality and video recommendations.	QA Team
System Testing	The entire YouTube web application will be tested as a complete, integrated system to validate its end-to-end functionality, performance, and overall compliance with business requirements.	QA Team

3.3 Test Types

Type	Objective	Technique	Tools
Functional Testing	To verify that all specified functionalities of the YouTube web application operate correctly according to requirements.	Requirements-based testing, User Story mapping.	Browser Developer Tools, Manual Test Cases
UI-UX Testing	To ensure the graphical user interface is intuitive, aesthetically pleasing, and provides a positive user experience across various devices and browsers.	Heuristic evaluation, Usability walkthroughs.	Browser Developer Tools (responsive design mode), Visual Inspection, Figma/Sketch (for design comparison)
Regression Testing	To confirm that recent code changes or additions have not adversely affected existing functionalities.	Risk-based regression, automated test suite execution.	Selenium WebDriver (for automation), Test Management Tool (e.g., Jira with Xray)

3.4 Test Design Techniques

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Use Case Testing

3.5 Execution Strategy

Test execution will primarily be manual for initial functional and UI-UX testing due to the complexity of the user interface and interactions. Automated test scripts will be developed for critical and stable functionalities. Tests will be prioritized based on feature criticality and risk, with high-priority tests executed first to ensure core functionality. Results will be meticulously recorded in the defect management system.

3.6 Regression Strategy

Regression testing will be conducted incrementally following each development sprint and prior to major releases. A prioritized suite of automated regression tests will be executed. Manual regression will be performed for areas identified as high-risk or prone to frequent changes. Full regression cycles will be scheduled for significant application updates.

3.7 Defect Management

Defects will be meticulously logged in a dedicated defect tracking system (e.g., Jira). Each defect will include a unique ID, detailed description, steps to reproduce, actual and expected results, severity, priority, and relevant attachments (screenshots, logs). Defects will be tracked through their lifecycle from 'New' to 'Closed' and re-tested upon resolution by the development team.

3.8 Test Data Strategy

Test data will be precisely managed to ensure realistic and comprehensive testing scenarios. Data will be provisioned from a combination of anonymized production data, synthetically generated data for specific edge cases, and manually created data for complex scenarios. Data integrity and privacy will be strictly maintained, aligning with organizational policies.

3.9 Environment Strategy

The dedicated QA environment will mirror the production environment in terms of software versions, configurations, and network settings as closely as possible. It will be configured to support various desktop and mobile device simulations across Chrome, Firefox, and Edge browsers. Environment stability and accessibility will be continuously monitored to prevent testing impediments.

4.0 Test Schedule & Resources

4.1 Milestones

Milestone	Target Date	Owner	Deliverable
Test Plan Review	2026-04-24	Heba Albaw	Approved Test Plan
Test Environment Setup	2026-04-29	Anna Liebert	Ready Test Environment
Test Case Review	2026-04-30	Johan Liebert	Reviewed Test Cases
Test Execution Start	2026-05-01	Heba Albaw	Initiated Test Cycle
Mid-Cycle Review	2026-05-04	Heba Albaw	Progress Report
Test Execution Complete	2026-05-08	Heba Albaw	Executed Test Cases
Bug Fix Verification	2026-05-12	Anna Liebert	Verified Bug Fixes
Final Report	2026-05-15	Heba Albaw	Final Test Report

4.2 Resource Plan

Role	Name	Responsibilities	Allocation
Test Manager	Heba Albaw	Overall test strategy and planning; Resource allocation and management; Stakeholder communication; Final report approval	100%
QA Engineer	Anna Liebert	Test environment setup and maintenance; Test case execution; Defect reporting and tracking; Regression testing	100%
QA Engineer	Johan Liebert	Test case design and review; Test data preparation; Test case execution; Performance testing support	100%

4.3 Effort Estimate

Activity	Hours
Test Planning	16
Test Design	24
Test Execution	72
Bug Reporting	16
Reporting	8

5.0 Risk Analysis

ID	Category	Description	Likelihood	Impact	Level	Mitigation
R-001	Environment	Unavailability of third-party services (e.g., content delivery networks, payment gateways, analytics platforms) critical for YouTube's functionality.	Medium	High	High	Identify all critical third-party dependencie Implement mock service or stubs for integration testing to isolate systen under test. Monitor third-party service status pages.
R-002	Resource	Test data becoming outdated or insufficient for comprehensive testing due to frequent application updates or data volume changes.	Medium	Medium	Medium	Establish a robust test data management strategy including automated data generation scripts and a refresh schedule. Utilize production-like data anonymized for testing.
R-003	Technical	Complexity of advanced UI/UX features (e.g., real-time video streaming, live chat, complex interactive components) leading to difficulties in automation or	Medium	High	High	Allocate dedicated time for researching and implementing advanced automation techniques. Involve UI/UX designers anc developers in

ID	Category	Description	Likelihood	Impact	Level	Mitigation
		accurate validation.				test case reviews to ensure accurate interpretation of requirements
R-004	Schedule	Tight testing schedule (2026-05-01 to 2026-05-08) may not allow for retesting of critical defects or full regression cycle if numerous issues are discovered.	Medium	High	High	Prioritize test cases based on business criticality and risk. Leverage existing automated regression suites. Implement daily stand-ups to monitor progress and identify roadblocks early.
R-005	Technical	Inaccurate or incomplete requirements for new features, particularly concerning edge cases or user interactions, impacting test coverage and defect detection.	Medium	Medium	Medium	Conduct thorough requirement reviews with business analysts and developers. Implement behavior-driven development (BDD) to clarify feature behavior.
R-006	Environment	Inconsistencies in performance test environment setup (e.g., b f	Medium	High	High	Regularly synchronize performance test environment fi ti

6.0 Entry & Exit Criteria

6.1 Entry Criteria

- All functional requirements and design specifications are formally approved and baselined.
- A stable, production-like test environment is provisioned, configured, and accessible to the test team.
- All necessary test data is generated, anonymized where required, and loaded into the test environment.
- The application build, suitable for testing, is successfully deployed to the test environment.
- All critical dependencies (e.g., third-party integrations, APIs, databases) are stable and available.
- Test cases for Functional, UI-UX, and Smoke testing are peer-reviewed and finalized.
- Required test tools (e.g., performance testing tools, defect tracking system) are installed, configured, and operational.

6.2 Exit Criteria

- All critical and major defects (P1, P2) for Functional, UI-UX, and Regression testing are resolved, re-tested, and closed.
- The test coverage for Functional and UI-UX test cases meets or exceeds the defined threshold (e.g., 95%).
- All Smoke tests pass successfully on the target build.
- Performance testing metrics meet or exceed the defined non-functional requirements (e.g., response times, throughput, resource utilization).
- Regression test cycles show no new critical or major defects introduced by recent changes.
- User Acceptance Testing (UAT) is successfully completed and signed off by stakeholders.
- All test artifacts, including test reports, defect logs, and traceability matrices, are complete and archived.

6.3 Suspension Criteria

- Discovery of a critical defect (P1) that renders a significant portion of the application untestable.
- Unavailability or instability of the test environment for more than four consecutive hours.
- A new build deployment introduces significant regressions, invalidating a major portion of completed testing.
- Non-compliance with security protocols or data privacy regulations identified within the test environment or application.
- Test data corruption or exhaustion prevents the execution of critical test scenarios.

6.4 Resumption Criteria

- The critical defect causing suspension is resolved, independently verified, and a new stable build is deployed.

- The test environment is restored to a stable and accessible state, and its stability is confirmed.
- The new build addressing regressions is successfully deployed, and initial smoke tests pass.
- Security and data privacy non-compliance issues are fully addressed and verified by the appropriate teams.
- Sufficient and valid test data is re-provisioned and loaded into the test environment.

7.0 Deliverables & Sign-off

7.1 Deliverables

Document	Description	Owner	Due	Format
Test Plan	Comprehensive document outlining the scope, approach, resources, and schedule of the testing activities for the video streaming platform.	Heba Albaw	2026-05-15	PDF
Test Cases	Detailed set of test cases, including preconditions, steps, expected results, and post-conditions for various functionalities of the video streaming platform.	Anna Liebert, Johan Liebert	2026-05-29	XLSX
Test Data	Collection of data sets required for executing the defined test cases, including user accounts, content metadata, and subscription configurations.	Anna Liebert, Johan Liebert	2026-05-29	CSV
Bug Reports	Detailed documentation of identified defects, including steps to reproduce, actual results, severity, and priority.	Anna Liebert, Johan Liebert	Ongoing	JIRA Ticket
Test Summary Report	Final report summarizing the testing activities, results, discovered defects, and overall assessment of the software quality.	Heba Albaw	2026-06-30	PDF
Lessons Learned	Document consolidating insights, challenges, and recommendations from the testing phase to improve future testing processes.	Heba Albaw	2026-07-07	DOCX

7.2 Approvals

Role	Name	Responsibility
QA Lead	Heba Albaw	Overall approval of the test plan and confirmation of adherence to quality standards.
Project Manager	To be determined	Approval of the test plan's alignment with project goals, schedule, and resource allocation.
Development Lead	To be determined	Review of the test plan for technical feasibility and agreement on testing scope and defect resolution

7.3 Testware Items

- Test Plan Document
- Test Cases Document
- Test Data Files
- Bug Reports (Jira tickets)
- Test Environment Configurations
- Test Automation Scripts
- Performance Test Scripts
- Security Test Scenarios
- Test Summary Report
- Lessons Learned Document